5. The Adjustment Criterion

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Adjustment Criterion* (Prof. Sanghack Lee, GSDS SNU)

The central question is exactly which covariate sets $\mathbf{Z}$ make the adjustment formula $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})P(\mathbf{z})$ equal to $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, and how such sets can be tested, constructed and enumerated efficiently.

These notes consolidate repeated slides, work every example in full (including the large twelve-variable example), and give the enumeration algorithm with its delay analysis. Physical PDF page numbers are listed in Section 7.

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1 Back-door criterion: recap and two refinements

Purpose. Recall back-door adjustment and see how it resolves Simpson's paradox and conditional effects.

Theorem 1.1 (Back-door adjustment; Pearl 2000, Thm. 3.3.2)**.** *If* $\mathbf{Z}$ *satisfies (i) no node in* $\mathbf{Z}$ *is a descendant of* $\mathbf{X}$*, and (ii)* $\mathbf{Z}$ *blocks every path between* $X \in \mathbf{X}$ *and* $Y \in \mathbf{Y}$ *containing an arrow into* X*, then* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.

1.1 Back to Simpson's paradox

Recall the drug/sex/recovery tables of Lecture 1 and the three stories:

	$F \to X$, $F \to Y$, $X \to Y$	$X \to F \to Y$, $X \to Y$	$X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$
$\emptyset$	not admissible	admissible	admissible
$\{F\}$	admissible	not admissible ($F \in De(X)$)	not admissible (opens collider)
Use	F-specific tables	aggregated table	aggregated table

The back-door criterion turns the physician's dilemma into a mechanical check on the graph.

1.2 Conditional back-door

Theorem 1.2 (Conditional back-door criterion)**.** *The* $\mathbf{w}$*-specific effect* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ *is identified whenever we can measure a set* $\mathbf{Z}$ *such that* $\mathbf{W} \cup \mathbf{Z}$ *satisfies the back-door criterion. Moreover,*

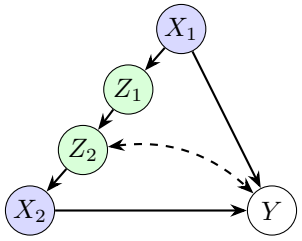
$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathbf{w}).$$

The derivation uses do-calculus (Lecture 6): Rule 2 exchanges $\mathrm{do}(\mathbf{x})$ with $\mathbf{x}$ given $\mathbf{Z} \cup \mathbf{W}$, and Rule 3 removes $\mathrm{do}(\mathbf{x})$ from $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$.

2 Tightness of the back-door criterion

Purpose. Show that the back-door criterion is sufficient but not necessary, and locate what must really be avoided.

Example 2.1. $\mathbf{X} = \{X_1, X_2\}$ with $X_1 \to Z_1 \to Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $Z_2 \leftrightarrow Y$.



Every variable other than $\mathbf{X}, Y$ descends from X_1, so no back-door admissible set exists. Yet with $\mathbf{Z} = \{Z_1, Z_2\}$,

$$P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(y \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

(Z_2 blocks $X_2 \leftarrow Z_2 \leftrightarrow Y$; conditioning on Z_2 opens the collider path $X_1 \to Z_1 \to Z_2 \leftrightarrow Y$, which Z_1 blocks.)

Condition (i) forbids descendants of $\mathbf{X}$ so as not to disturb the causal paths. But some descendants of $\mathbf{X}$ may be used, and sometimes are needed, for adjustment: in particular those that are also ancestors of $\mathbf{X}$ and do not disturb causal paths. *Which variables must really be avoided?*

3 The adjustment criterion

Purpose. Characterize all valid adjustment sets (necessary and sufficient conditions).

So far: $\mathbf{Z}$ back-door w.r.t. $\langle \mathbf{X}, \mathbf{Y} \rangle \implies$ adjustment formula. Now we ask for a condition "???" with

$$\mathbf{Z} \text{ is ??? w.r.t. } \langle \mathbf{X}, \mathbf{Y} \rangle \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

3.1 Proper causal paths

Definition 3.1 (Proper causal path; Shpitser et al. 2010)**.** A path from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$ is *causal* if it is directed, and *proper* if it contains a member of $\mathbf{X}$ only at its starting point.

Example 3.2. $X_1 \to W_1 \to X_2 \to W_2 \to Y$, $X_1 \to W_3 \to Y$, $X_1 \leftrightarrow Y$. Proper causal paths: $X_1 \to W_3 \to Y$ and $X_2 \to W_2 \to Y$. Non-proper: $X_1 \to W_1 \to X_2 \to W_2 \to Y$ (it passes through X_2).

3.2 Disturbing a proper causal path

A proper causal path is *disturbed* by adjusting for a variable on the path (other than $\mathbf{X}$) or for a descendant of such a variable.

Example 3.3. $X \to W \to Y$, $W \to Z$. All X–Y paths are causal, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$ (back-door with $\emptyset$). Adjusting for Z, a descendant of W on the proper causal path:

$$\sum_z P(y \mid x, z) P(z) = \sum_{z,w} P(y \mid x, z, w) P(w \mid x, z) P(z) = \sum_w P(y \mid w) \sum_z P(w \mid x, z) P(z),$$

$$P(y \mid \mathrm{do}(x)) = P(y \mid x) = \sum_{w,z} P(y \mid x, z, w) P(w \mid x, z) P(z \mid x) = \sum_w P(y \mid w) \sum_z P(w \mid x, z) P(z \mid x).$$

For SCMs agreeing on $P(y \mid w)$ and $P(w \mid x, z)$ but with $P(z) \neq P(z \mid x)$ (which is generic, since Z depends on X through W), the two expressions differ. The same happens if we adjust for W itself.

3.3 The criterion

Theorem 3.4 (Adjustment criterion; Shpitser, VanderWeele and Robins 2010)**.** $\mathbf{Z}$ *satisfies the adjustment criterion relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$ *iff*

(i) $\mathbf{Z}$ *does not disturb any proper causal path: no element of* $\mathbf{Z}$ *is a descendant, in* $\mathcal{G}_{\overline{\mathbf{X}}}$*, of any* $W \notin \mathbf{X}$ *lying on a proper causal path from* $\mathbf{X}$ *to* $\mathbf{Y}$*; and*

(ii) all proper non-causal paths in $\mathcal{G}$ *from* $\mathbf{X}$ *to* $\mathbf{Y}$ *are blocked by* $\mathbf{Z}$*.*

$\mathbf{Z}$ *satisfies the adjustment criterion* if and only if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$ *for every model compatible with* $\mathcal{G}$.

The proof is too involved for class. Sufficiency can be obtained from do-calculus; necessity constructs, for any violating $\mathbf{Z}$, a model in which the formula fails.

3.4 Implementation idea

Input: $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathcal{G}$. **Output:** whether $\mathbf{Z}$ is adjustment-admissible.

Step 1 (proper causal paths). Compute the vertices on proper causal paths,

$$\mathrm{PCP} = (De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \setminus \mathbf{X}) \cap An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}},$$

descendants of $\mathbf{X}$ (after removing edges into $\mathbf{X}$) that reach $\mathbf{Y}$ without passing through $\mathbf{X}$ again. (The slides use the simpler superset $\mathbf{W} = De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, and $\mathcal{H} = \mathcal{G}[\mathbf{W}]$ with bidirected edges removed to capture the PCP structure; computing ancestors in $\mathcal{G}_{\underline{\mathbf{X}}}$ removes vertices that reach $\mathbf{Y}$ only through another treatment.)

Step 2 (forbidden variables). $\mathcal{F} = De(\mathrm{PCP})_{\mathcal{G}_{\overline{\mathbf{X}}}}$: variables on proper causal paths (excluding $\mathbf{X}$) and their descendants. Conditioning on any of them could block a causal path or create bias. *Check (i):* if $\mathbf{Z} \cap \mathcal{F} \neq \emptyset$, $\mathbf{Z}$ is not admissible; stop.

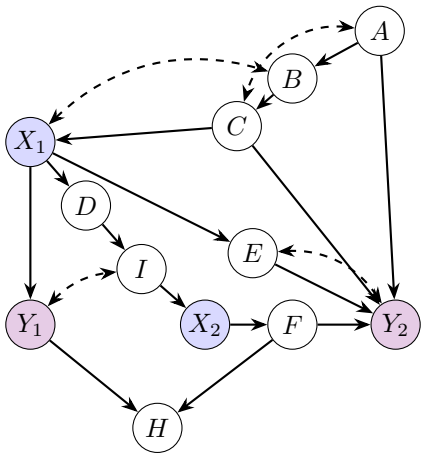
Step 3 (proper non-causal paths). Build $\mathcal{G}'$ by removing the first edge of every proper causal path,

$$\mathcal{G}' = \mathcal{G} \setminus \{X \to W : X \in \mathbf{X},\ W \in \mathrm{PCP}\}.$$

In $\mathcal{G}'$ all causal paths are cut, so only non-causal paths remain. *Check (ii):* $\mathbf{Z}$ is admissible iff $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}'}$.

Summary: find PCPs $\to$ check $\mathbf{Z} \cap \mathcal{F} = \emptyset$ $\to$ check d-separation in $\mathcal{G}'$. Each step is a graph search, so the test is polynomial.

3.5 An example



$\mathcal{G}$: $A \to B \to C \to X_1$, $C \to Y_2$, $A \to Y_2$, $X_1 \to D \to I \to X_2$, $X_1 \to E \to Y_2$, $X_1 \to Y_1$, $X_2 \to F \to Y_2$, $Y_1 \to H \leftarrow F$, and bidirected $X_1 \leftrightarrow B$, $A \leftrightarrow C$, $Y_1 \leftrightarrow I$, $E \leftrightarrow Y_2$; $\mathbf{X} = \{X_1, X_2\}$, $\mathbf{Y} = \{Y_1, Y_2\}$.

(i) Forbidden set. Proper causal paths: $X_1 \to Y_1$, $X_1 \to E \to Y_2$, $X_2 \to F \to Y_2$. ($X_1 \to D \to I \to X_2 \to \cdots$ is not proper.) Their non-treatment vertices are $\{Y_1, E, Y_2, F\}$, whose descendants are $\{Y_1, E, Y_2, F, H\}$; apart from $\mathbf{Y}$, $\mathcal{F} = \{E, F, H\}$.

(ii) Evaluating $\mathbf{Z} = \{C, D, I\}$. $\mathbf{Z} \cap \mathcal{F} = \emptyset$, so (i) holds. Cut the first edges $X_1 \to Y_1$, $X_1 \to E$, $X_2 \to F$ to obtain $\mathcal{G}'$. Any remaining path between $\mathbf{X}$ and $\mathbf{Y}$ is non-causal and must be blocked by $\mathbf{Z}$ (with no E, F, H). But $X_1 \leftrightarrow B \leftarrow A \to Y_2$ is open: B is a collider and an ancestor of $C \in \mathbf{Z}$, and A is an unobserved non-collider. Hence $(\mathbf{X} \not\perp_d \mathbf{Y} \mid C, D, I)_{\mathcal{G}'}$ and $\mathbf{Z}$ is *not* admissible. A path that is non-causal in $\mathcal{G}'$ corresponds to a proper non-causal path in $\mathcal{G}$.

3.6 Explicit construction of an admissible set

Lemma 3.5 (Explicit admissible set)**.** *Let $\mathcal{F}$ be the variables forbidden by condition (i) and $An^-(\mathbf{W}) = An(\mathbf{W}) \setminus \mathbf{W}$. If any admissible set exists, then*

$$\mathbf{Z}_0 = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$$

is admissible. (Ancestors should be taken in $\mathcal{G}'$, but the result is the same in $\mathcal{G}$.)

In the example, $An^-(\mathbf{X} \cup \mathbf{Y}) = \{A, B, C, D, E, F, I\}$ and $\mathbf{Z}_0 = \{A, B, C, D, I\}$. Now $A \in \mathbf{Z}_0$ blocks $X_1 \leftrightarrow B \leftarrow A \to Y_2$ and $X_1 \leftrightarrow B \leftarrow A \leftrightarrow C \to Y_2$; C blocks $X_1 \leftarrow C \to Y_2$; D blocks $X_1 \to D \to I \leftrightarrow Y_1$ (the collider I is opened by conditioning); I blocks $X_2 \leftarrow I \leftrightarrow Y_1$. Hence $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z}_0)_{\mathcal{G}'}$ and $\mathbf{Z}_0$ is admissible. Note that D and I are descendants of X_1: the back-door criterion would never allow them.

Testing existence. If $\mathbf{Z}_0$ is not admissible, no admissible set exists: a *one-shot* verification of the existence of a solution.

3.7 Recap

- The adjustment criterion characterizes all valid adjustment sets (necessary and sufficient).
- Condition (i): $\mathbf{Z}$ contains no variable of $\mathcal{F}$ (descendants of variables on proper causal paths).
- Condition (ii): $\mathbf{Z}$ blocks all proper non-causal paths from $\mathbf{X}$ to $\mathbf{Y}$.
- Testing and constructing admissible sets takes polynomial time.

4 Listing all admissible sets (optional)

Purpose. Enumerate every admissible set with polynomial delay, so that sets can be chosen by cost, variance or other preferences.

4.1 Motivation

There are typically several admissible sets. Which one to use may depend on availability, measurement cost, variance, privacy or fairness. Example: $A \to X$, $A \to B \to Y$, $X \to C \to Y$. The sets $\{A, B\}$, $\{A\}$ and $\{B\}$ are all admissible. Which should we prefer? What if A is a "genetic condition" (expensive or sensitive) and B is "headache"?

4.2 Challenge: output size

Even with an oracle f returning all solutions in $O(1)$, the loop `for Zs in f(G, Xs, Ys): output(Zs)` can take exponential time, because the number of solutions can be exponential. With k parallel paths $X \to W_{i,1} \to W_{i,2} \to Y$, a set separating X from Y must contain $W_{i,1}$, $W_{i,2}$ or both for each i: 3^k separators. So total running time is the wrong yardstick.

Definition 4.1 (Polynomial delay)**.** A listing algorithm runs with *polynomial delay* if there is a constant d such that it spends at most $O(n^d)$ time (1) between the start and either producing the first output or halting with no output, and (2) between producing one output and either producing the next or halting.

4.3 Theoretical tool: bounded separators

Lemma 4.2 (Inclusion and restriction)**.** *Given $\mathcal{G}$ and disjoint $\mathbf{X}, \mathbf{Y}$, and $\mathbf{I} \subseteq \mathbf{R} \subseteq \mathbf{V}$: if any set $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}$, then*

$$\mathbf{Z}_0 = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I}) \cap \mathbf{R}$$

also separates them.

$\mathbf{I}$ is the "include" set and $\mathbf{R}$ the "restrict to" set. Existence of a separator between the bounds is checked by testing $\mathbf{Z}_0$, in $O(n+m)$ time; call this `exist-sep`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$. With $\mathbf{I} = \emptyset$ and $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ this is the explicit construction above.

4.4 Algorithm

Divide and conquer: implement `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$, outputting all separators $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$. Let $\mathcal{F}^+ = \mathbf{X} \cup \mathbf{Y} \cup \mathcal{F}$. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$, and use $\mathcal{G}'$ ($\mathcal{G}$ with the first edge of each proper causal path removed). Every output separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition (ii)) and avoids $\mathcal{F}^+$ (condition (i)), so every output satisfies the adjustment criterion.

Algorithm 1 `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$

```
if exist-sep(G, X, Y, I, R) then                          ▷ O(n + m)
    if I = R then output I
    else
        pick W ∈ R \ I
        list-seps(G, X, Y, I ∪ {W}, R)                    ▷ include W
        list-seps(G, X, Y, I, R \ {W})                    ▷ exclude W
```

Example 4.3 (Running example)**.** $A \to X$, $A \to B \to Y$, $D \to X$, $D \leftrightarrow Y$, $X \to C \to Y$.

1. Forbidden: variables on proper causal paths other than X and their descendants, plus X, Y: $\mathcal{F}^+ = \{C, X, Y\}$.
2. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \{A, B, D\}$, and remove $X \to C$.
3. Branch on D. Excluding D ($\mathbf{R} = \{A, B\}$) is pruned immediately: $X \leftarrow D \leftrightarrow Y$ cannot be blocked. Including D, branch on A, then on B. Leaves: $\{A, B, D\}$ (output), $\{A, D\}$ (output), $\{B, D\}$ (output), $\{D\}$ (pruned: $X \leftarrow A \to B \to Y$ open).

The admissible sets are $\{A, B, D\}, \{A, D\}, \{B, D\}$.

4.5 Analysis

- A node of the recursion tree is expanded only if `exist-sep` is true; otherwise no separator lies between $\mathbf{I}$ and $\mathbf{R}$ and the branch is safely ignored. Hence every expanded node has at least one output leaf below it.

- Every reached leaf produces an output $\mathbf{Z} = \mathbf{I} = \mathbf{R}$, which separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition ii) and, since $\mathbf{R}$ initially excludes $\mathcal{F}^+$, satisfies condition (i).
- The tree depth is at most $k = n - |\mathcal{F}^+| < n$, so at most $O(n)$ recursive calls separate two consecutive outputs.
- Each call costs $O(n + m)$ (`exist-sep`), $O(1)$ for the comparison and $O(n)$ to output. The delay is therefore $O(n(n + m))$.

The total time may still be exponential (unavoidable), but outputs arrive one after another in polynomial time, and the algorithm halts in polynomial time if there are no solutions.

5 Connection with conditional ignorability

Purpose. Relate the graphical criterion to the key assumption of the potential-outcome framework.

The central identifying assumption of the potential outcome framework is *conditional ignorability*,

$$\mathbf{Y}_{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z},$$

written with counterfactuals. It requires that, conditional on $\mathbf{Z}$, the treatment is independent of the outcomes under intervention, which gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$ and hence the adjustment formula. In the PO framework this is usually *assumed*; the adjustment criterion *justifies* it by stating under which conditions such a set exists given a model of reality.

6 Summary

- Adjustment is arguably the most popular technique across the data sciences for identifying causal effects.
- The back-door/adjustment criterion gives a principled condition for the validity of adjusting for $\mathbf{Z}$.
- Admissible sets (if any) can be found systematically and efficiently given the graph and the target effect.
- Once a set is found, estimation techniques (IPW and variants) evaluate the effect efficiently and robustly from finite samples.

7 Source coverage

The source deck has 70 physical pages (33 numbered slides with build-ups).

Topic	Physical pages
Back-door recap, Simpson revisited, conditional back-door	1–7
Tightness of the back-door criterion	8–12
Adjustment criterion: PCPs, disturbance, definition, implementation	13–27
Worked example and explicit construction	28–40
Listing algorithm: motivation, delay, tool, recursion, analysis	41–61
Conditional ignorability, summary, references	62–70